TCP for Wireless Environment

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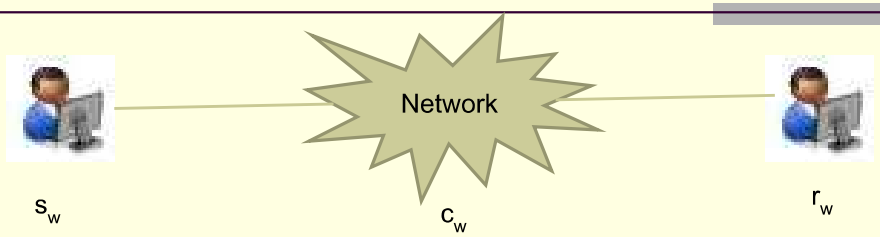
TCP Congestion Control

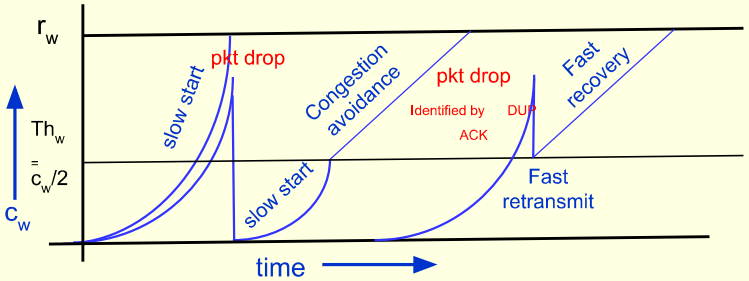
- Slow Start
- Congestion Avoidance
- Fast Retransmit
- Fast Recovery

TCP Congestion Control



Speed = min(s_w, c_w, r_w)

Sliding window = min(c_w, r_w)



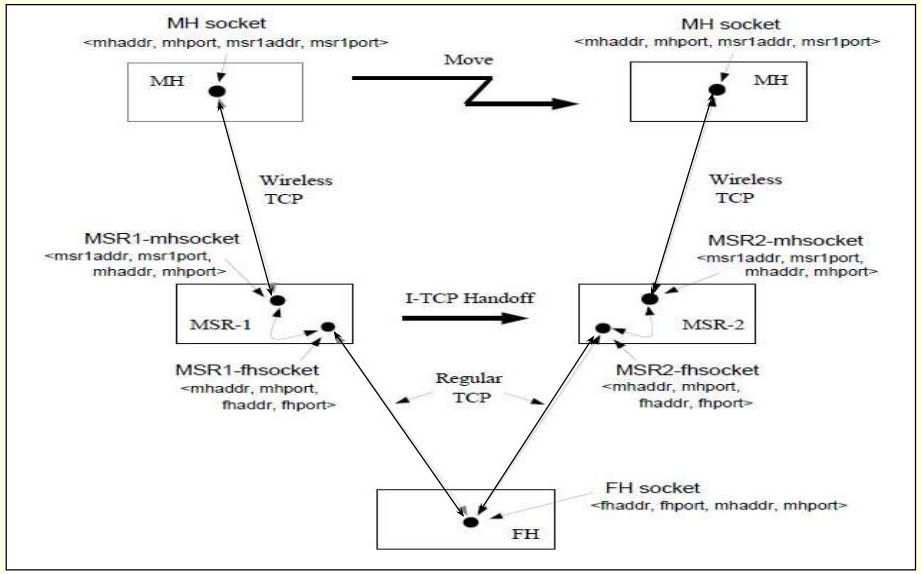
Factors affecting TCP performance in Wireless Environment

- Limited Bandwidth
  - Typically 11Mbps on WLAN as compared to 100 Mbps on LAN
- Long Round Trip Time
  - Increment in congestion window is directly proportional to the rate of received ACKs.
- Packet Loss at last hop
  - The loss of packets in wireless link, which in general is the last hop, results in end-to-end retransmission.
- User Mobility
  - TCP attributes delays and losses caused by these short periods of disconnection to congestion and triggers congestion control and avoidance mechanism.
- Short Flows
  - There is a high probability that the transfer completes even before the sender's window reaches the maximum possible size.

Indirect TCP (I-TCP)

- ***Segmentation*** of TCP connection into a fixed part & a wireless part
  - Segmentation point is the access point like Foreign Agent in case of MobileIP or a Base Station in case of WLAN
- The access point (FA/BS) acts as a ***proxy***.
  - Standard TCP is used for the wired link and a special TCP is adapted for the wireless link.
  - When the FH sends a packet, the access point acknowledges it, buffers it & try to forward the buffered packets to the MH.
    - ✔ The buffer will be cleared when the MH acknowledges the packet.
  - Similarly, when MH sends a packet, the access point acknowledges it, buffers it & try to forward the buffered packets to the FH

I-TCP (Cont.)



I-TCP (Cont.)

- Advantages:
  - *Doesn't require any change in TCP Protocol for fixed network.*
  - *High error rates in wireless link is not propagated to the fixed network.*
  - *A different transport layer protocol may be incorporated for wireless network.*
- Disadvantages:
  - *The end-to-end semantics (e.g. reliability) is not guaranteed*
  - *Difficulty of buffer management during handover.*
  - *Incorporating end-to-end security requires trusted access points.*

Snooping TCP (S-TCP)

- Two modules at the access point

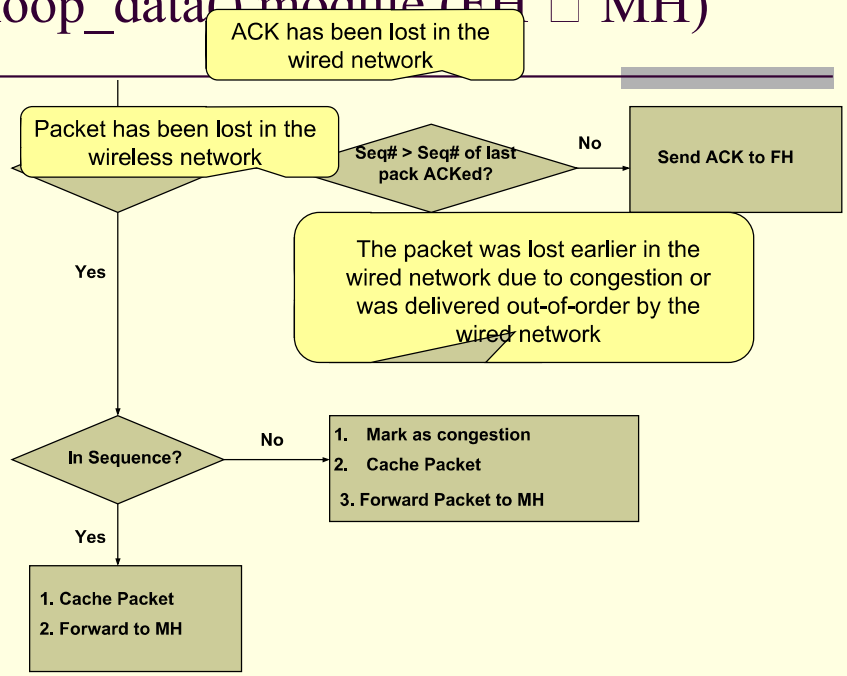
  Snoop_data()

  Processes data packets from the FH as well as MH

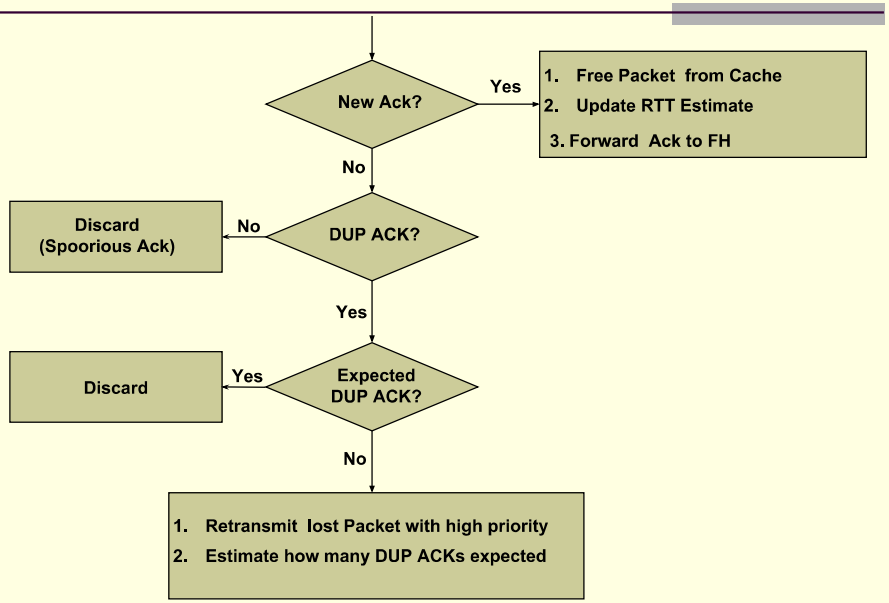
  Snoop_ack()

  Monitors acks sent by MH

Snoop_data() module (FH □ MH)



Snoop_ack() module (MH □ FH)



Snoop_data() module (MH □ FH)

- Access point monitors data packets from MH
- Access point uses sequence numbers to detect lost packets and send NACKs for the lost ones.
- NACKs implemented using SACK (Selective Ack)

S-TCP (Cont.)

- Advantages:
  - *The end-to-end TCP semantics is preserved.*
  - *Enhancement is incorporated only in the access point. FH & MH need not be changed.*
  - *Buffers need not be moved during handover.*
  - *Every access point doesn't require the enhancement.*
- Disadvantages:
  - *The quality of isolation strongly depends on the time-out values.*
  - *Can't incorporate end-to-end security.*

Mobile TCP (M-TCP)

- Combines approaches from I-TCP & S-TCP.
- Better handling of disconnection.
- Like I-TCP, it splits the TCP connection into two parts at the access point.
  - *The access point is called a Supervisory Host (SH).*
  - *However unlike I-TCP, SH doesn't perform any buffering / retransmission of data.*
- Like S-TCP, SH monitors all packets send to the MH & ACKs returned by the MH.
  - When SH doesn't receive any ACKs for a specific period of time it assumes that the MH is disconnected.
    - *It then sets the sender's window size to 0 so that sender will not transmit any data.*
    - *As soon as SH detects connectivity again, it reopens the window to the sender's old value.*

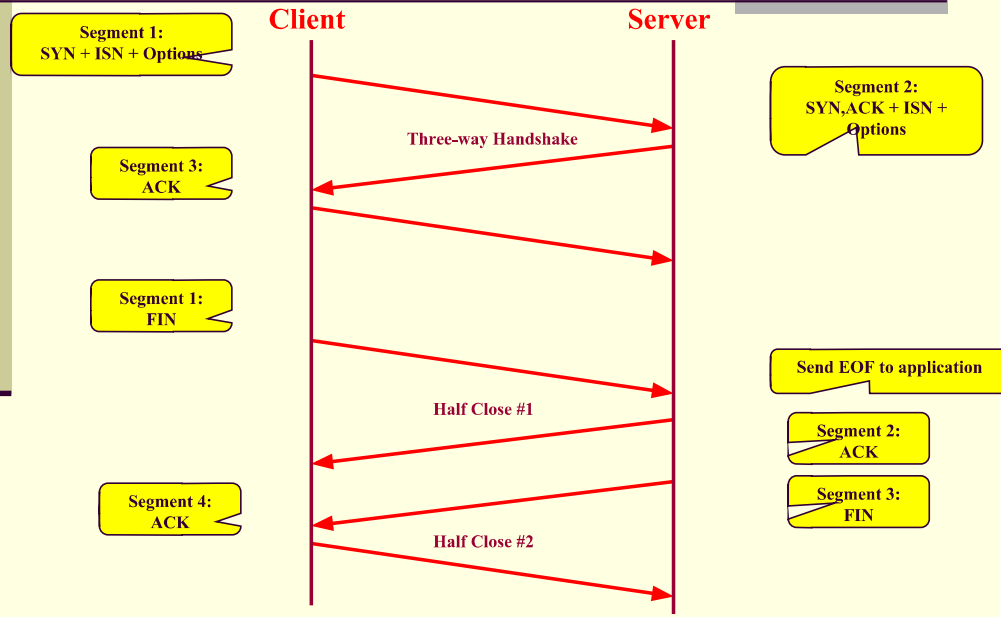
M-TCP (Cont.)

- Advantages:
  - *Maintains end-to-end TCP semantics.*
  - *Maintains end-to-end security.*
  - *When the MH disconnects, it avoids useless retransmission and congestion control.*
  - *Easy handovers.*
- Disadvantages:
  - *Packet loss is propagated to the sender.*

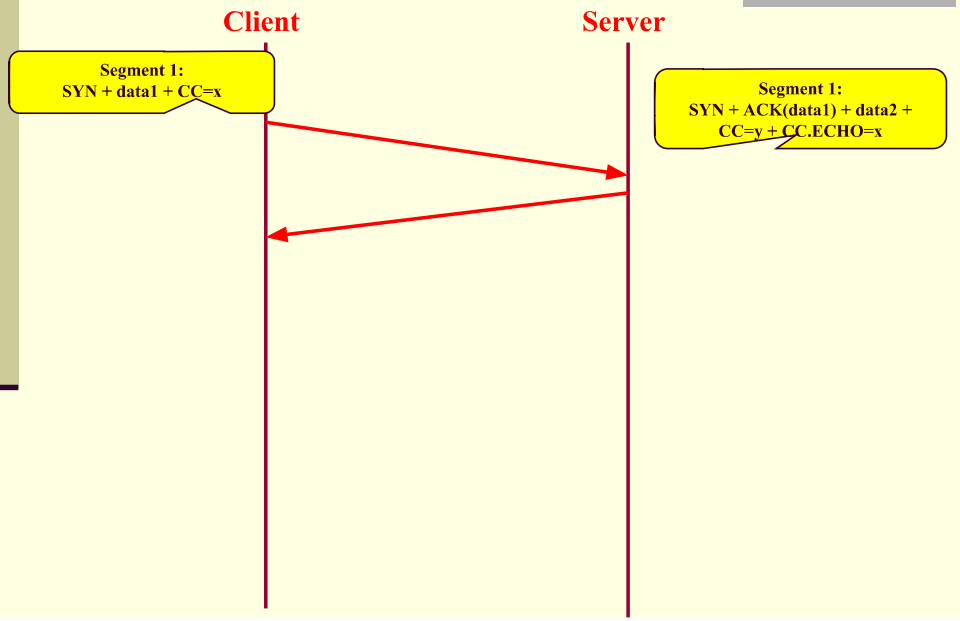
T-TCP

- TCP phases
  - connection setup, data transmission, connection release
  - using 3-way-handshake needs 3 packets for setup and 4 packets for release, respectively
  - thus, even short messages need a minimum of 8 packets!
- Transaction oriented TCP
  - RFC1644, T-TCP, describes a TCP version to avoid this overhead
  - connection setup, data transfer and connection release can be combined
  - thus, only 2 or 3 packets are needed

Normal TCP



T-TCP (Cont.)


Client
Server
Segment 1:
SYN + data1 + CC=x
Segment 1:
SYN + ACK(data1) + data2 +
CC=y + CC.ECHO=x